

Automated Warehouse Sorting Robot

Course: Artificial Intelligence
Assignment: PEAS Framework and Task Environment Analysis
Reference: Russell & Norvig – Artificial Intelligence: A Modern Approach, Chapter 2
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Introduction

For this assignment I picked the Automated Warehouse Sorting Robot, modelled on systems deployed by companies such as Amazon Robotics. Every time a customer places an online order, a robot like this one navigates the warehouse floor, locates the correct item on a storage shelf, picks it up, and transports it to a packing station so the order can be shipped. The entire process happens without direct human involvement for each individual item.

I selected this system because it is one of the most visible and impressive examples of an intelligent agent operating in a physical environment. I have seen videos of Amazon warehouses where hundreds of robots move around at the same time without crashing into each other, and I wanted to understand how the system is designed to make that possible.

The robot has to navigate a busy shared space filled with other robots and human workers, identify and carefully pick items of many different shapes and sizes, and do all of this quickly and accurately enough to keep up with the enormous volume of orders the warehouse processes every day. This makes it a strong example of an intelligent agent with clear goals, inputs, outputs, and performance measures.

Task 1: PEAS Framework

The system I am analyzing is an Automated Warehouse Sorting Robot deployed inside large fulfilment centres to locate, pick, and transport inventory items from storage shelves to packing and dispatch stations in real time.

Performance Measures

These are the things the system tries to do well.

1. Sorting and Picking Accuracy

This measures the percentage of pick attempts in which the robot correctly identifies the right item, grasps it successfully, and places it in the correct destination bin or tote. A mature system should achieve above 99% accuracy because even small error rates at the scale of millions of picks per day result in a large number of wrong orders reaching customers.

2. Throughput (Items Processed Per Hour)

This measures how many items the robot completes per shift. During peak demand periods such as major sales events, the warehouse must process a very high number of orders and the robot must maintain a sufficient throughput rate to prevent backlogs from forming in the packing and dispatch pipeline.

3. Damage and Mis-sort Rate

This tracks the share of items that are dropped, crushed, or placed in the wrong destination bin. Damaged items cannot be sold and must be written off as losses. Mis-sorted items cause incorrect orders to be shipped to customers, resulting in returns, refunds, and damage to customer trust. Keeping this rate as low as possible is essential for profitability.

4. Energy and Uptime Efficiency

This measures battery consumption per item moved and the proportion of scheduled working time the robot spends actively completing tasks rather than idling, queuing, or waiting at a charging dock. Since warehouses operate fleets of thousands of robots, even a small improvement in energy efficiency compounds into significant cost savings across the entire operation.

Environment

The Automated Warehouse Sorting Robot operates inside large physical fulfilment centres that run continuously twenty-four hours a day. The warehouse floor contains storage racks and pods holding thousands of different products, conveyor belts, packing stations, charging docks, and safety zones around human workstations.

The environment includes hundreds of other robots, human pickers and packers, a central fleet management computer, and a constantly changing queue of incoming customer orders. The physical layout of storage pods is rearranged regularly based on demand forecasts, meaning the robot cannot rely on a fixed map. Floor conditions such as surface texture and aisle clearance also vary. Wireless network connectivity is essential for the robot to receive task assignments, and any interruption

creates gaps in coordination.

Because of these factors, the robot must operate in a complex, physical, and constantly changing environment that is shared with many other agents simultaneously.

Actuators

These are the ways the robot performs actions and affects its environment.

- **Robotic Arm and End-Effector Assembly**

Uses interchangeable suction cups, pincer grippers, or soft robotic fingers to grasp, lift, and place items of different shapes, sizes, and weights with precise control.

- **Mobile Drive Base**

Uses omnidirectional wheels to navigate the warehouse floor, position the robot in front of shelving units, and dock with conveyors or charging stations.

- **Vertical Lift and Telescoping Mechanism**

Raises or lowers the arm assembly to reach items stored at any shelf height, allowing the robot to access the full vertical range of storage without human assistance.

- **Indicator and Signalling System**

Uses LED light arrays, audible tones, and on-screen status displays to communicate the robot's intended path and actions to nearby human workers, reducing the risk of collisions in a shared workspace.

Sensors

These are the inputs used by the robot to understand and evaluate its surroundings.

- **Barcode and RFID Scanners**

Provide direct confirmation of item identity by reading product labels and tags, ensuring the robot picks the correct item before transporting it.

- **Force, Torque, and Load-Cell Sensors**

Installed in the gripper to detect grip security and item weight. A sudden change in measured weight indicates an item is slipping, allowing the robot to correct its grip before dropping the item.

- **High-Resolution RGB-D Cameras**

Capture both colour and depth information to support grasp planning, shelf-slot localisation, and obstacle identification around the robot.

- **LIDAR and Ultrasonic Proximity Sensors**

Build and continuously update a real-time map of the surrounding aisle, detecting static obstacles as well as moving entities such as other robots and human workers.

- **Wheel Encoders and Inertial Measurement Unit (IMU)**

Track the robot's position, orientation, and velocity, providing navigation data that is combined with the facility's digital map for precise movement.

- **Fleet-Management and Warehouse-System Signals**

Receive live task assignments, congestion data from other robots, battery charge schedules, and order priority updates from the central warehouse management system.

Task 2: Environment Classification

Dimension	Classification	Justification
Fully Observable vs. Partially Observable	Partially Observable	The robot's onboard sensors cover only the area immediately around it, so shelving units, stacked totes, and other robots constantly block its view of the wider warehouse state.
Single-Agent vs. Multi-Agent	Multi-Agent	Hundreds of robots and human workers share the same aisles and charging infrastructure simultaneously, and the fleet management system must continuously arbitrate competing requests for the same physical space.
Deterministic vs. Stochastic	Stochastic	Even a previously successful grasp strategy on an identical item can fail if the item has shifted inside its packaging or has been placed at a slightly different angle, so outcomes cannot be predicted with certainty.
Episodic vs. Sequential	Sequential	Each pick and route decision changes the robot's battery level and the congestion state of the aisles, which directly affects the options available for future tasks.
Static vs. Dynamic	Dynamic	Other robots reposition, new orders arrive, and human workers move through the same aisles continuously while the robot is planning, so the warehouse is never frozen between decisions.
Discrete vs. Continuous	Continuous	Arm joint angles, gripper force, wheel velocities, and obstacle distances are all real-valued signals measured on continuous numerical scales.

Task 3: Critical Analysis

3.1 Most Technically Challenging Dimension

In my opinion the hardest part of building this system is dealing with the Dynamic Environment. The warehouse never stops changing while the robot is working. Other robots keep moving through the same aisles, human workers walk unpredictably across the floor, and new orders arrive constantly and change which items need to be picked next.

The challenge is that every path and action plan the robot creates is based on the state of the warehouse at that specific moment. By the time the robot starts moving, another robot may have entered the same aisle, or a human worker may have stepped into the planned path. This means the robot cannot simply plan once and then follow the plan. It has to continuously re-plan its movements while it is already in motion, which requires very fast processing and constant coordination with all the other robots in the fleet.

Another difficulty is preventing deadlocks, which happen when two or more robots block each other's only available path and neither one can move forward. Detecting and resolving these situations before they bring an entire section of the warehouse to a stop is still an active area of research in warehouse robotics.

Because of these factors, maintaining performance in a dynamic and shared physical environment remains the most technically demanding aspect of warehouse robot engineering.

3.2 Utility Function and Trade-Off Analysis

For an Automated Warehouse Sorting Robot the main trade-off exists between completing as many picks as possible quickly and doing so without damaging items or causing safety incidents involving other robots or human workers.

The proposed utility function is:

$$U = \alpha \times \text{Throughput_Score} - \beta \times \text{Damage_Risk} - \gamma \times \text{Collision_Risk}$$

Where Throughput Score represents the number of items successfully picked and transported per hour, Damage Risk represents the estimated probability of dropping or crushing an item during the current pick or transport manoeuvre, and Collision Risk represents the probability of a near-miss or contact with another robot or a human worker based on current speed and proximity.

If the weight of Throughput Score is increased significantly, the robot becomes much more aggressive in its movements. It selects faster travel speeds, tighter turns through aisles, and quicker grasping actions. This increases the number of items processed per hour in the short term but also raises the likelihood of dropping fragile items and getting dangerously close to other robots and human workers.

This demonstrates why warehouse robots must carefully balance productivity, item safety, and workplace safety. Optimizing one factor too aggressively can seriously harm the others and ultimately reduce the overall performance and safety of the entire warehouse operation.

Task 4: Structured JSON Representation

```
{
  "system_name": "Automated Warehouse Sorting Robot",
  "peas": {
    "performance_measures": [
      "Sorting and Picking Accuracy: percentage of pick attempts where the
        correct item is grasped and placed in the correct destination",
```

```

    "Throughput: number of items successfully processed per hour",
    "Damage and Mis-sort Rate: share of items dropped, crushed, or
    placed in the wrong destination bin",
    "Energy and Uptime Efficiency: battery consumption per item moved
    and proportion of time spent actively completing tasks"
  ],
  "environment": "Large physical warehouse fulfilment centre containing
  storage racks, conveyor belts, packing stations, charging docks,
  hundreds of co-located robots, human workers, and a continuously
  changing queue of customer orders.",
  "actuators": [
    "Robotic Arm and End-Effector Assembly: grasps and places items
    using suction cups, pincer grippers, or soft robotic fingers",
    "Mobile Drive Base: navigates the warehouse floor using
    omnidirectional wheels",
    "Vertical Lift and Telescoping Mechanism: raises or lowers the
    arm to access items at any shelf height",
    "Indicator and Signalling System: communicates intended movements
    to nearby human workers using LED lights and audible tones"
  ],
  "sensors": [
    "Barcode and RFID Scanners: confirm correct item identity before
    picking",
    "Force, Torque, and Load-Cell Sensors: detect grip security and
    weight changes indicating a slipping item",
    "High-Resolution RGB-D Cameras: capture colour and depth for
    grasp planning and obstacle identification",
    "LIDAR and Ultrasonic Proximity Sensors: build real-time maps
    of the surrounding aisle and detect nearby robots and people",
    "Wheel Encoders and IMU: track the robot's position, orientation,
    and velocity for precise navigation",
    "Fleet-Management and Warehouse-System Signals: receive live task
    assignments and congestion data from the central system"
  ]
},
"environment_classification": {
  "fully_observable_vs_partially_observable": {
    "choice": "Partially Observable",
    "justification": "Onboard sensors cover only the local area around
    the robot. Shelves and other robots block the wider warehouse view."
  },
  "single_agent_vs_multi_agent": {
    "choice": "Multi-Agent",
    "justification": "Hundreds of robots and human workers share the
    same aisles and infrastructure simultaneously."
  },
  "deterministic_vs_stochastic": {
    "choice": "Stochastic",
    "justification": "Grasp outcomes depend on unpredictable item
    position and packaging variation that the robot cannot observe."
  },
  "episodic_vs_sequential": {
    "choice": "Sequential",
    "justification": "Each pick and route decision changes battery
    level and aisle congestion, affecting all future task options."
  }
}

```

```
    },
    "static_vs_dynamic": {
      "choice": "Dynamic",
      "justification": "Robots, orders, and workers continuously change
the warehouse state while the robot is operating."
    },
    "discrete_vs_continuous": {
      "choice": "Continuous",
      "justification": "Arm joint angles, gripper force, velocities, and
obstacle distances are all continuous numerical values."
    }
  },
  "utility_function":
    "U = alpha * Throughput_Score - beta * Damage_Risk - gamma * Collision_Risk"
}
```

References

- Russell, S., & Norvig, P. (2022). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.
- Guizzo, E. (2008). Three Engineers, Hundreds of Robots, One Warehouse. *IEEE Spectrum*, 45(7), 26–34.
- Wurman, P. R., D’Andrea, R., & Mountz, M. (2008). Coordinating Hundreds of Cooperative, Autonomous Vehicles in Warehouses. *AI Magazine*, 29(1), 9–20.